

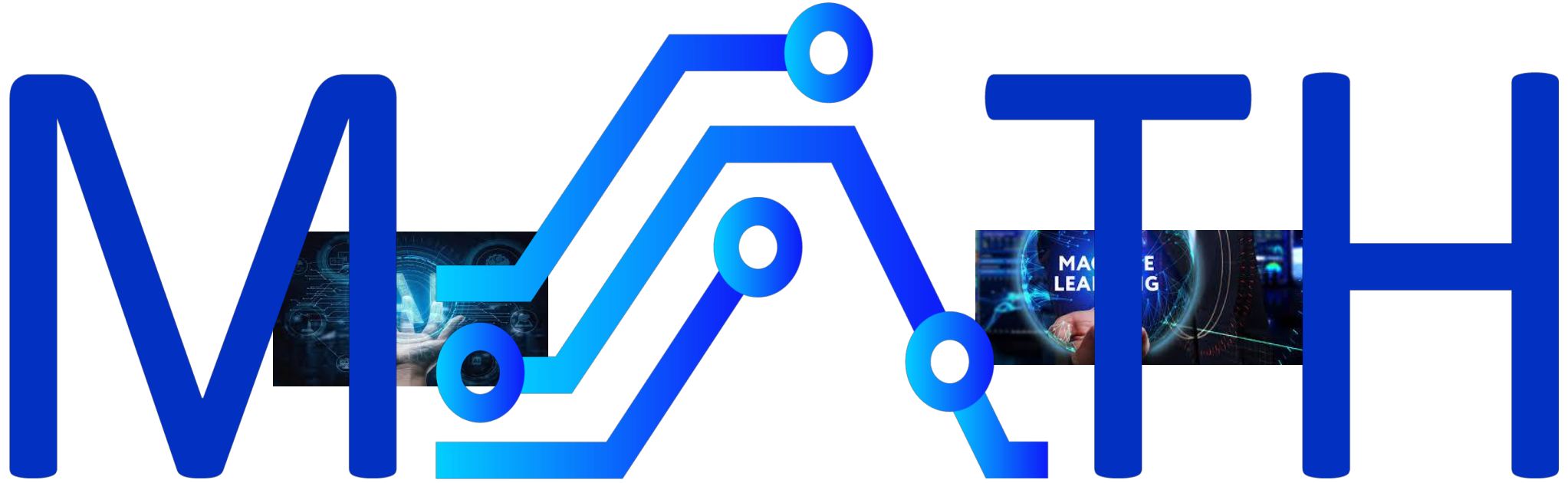


**TOPIC : SYLLABUS DISCUSSION
&
BASIC UNDERSTANDING (AI,ML,DL&DS)**

Presented by:
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(Assistant Professor, Quad AI)



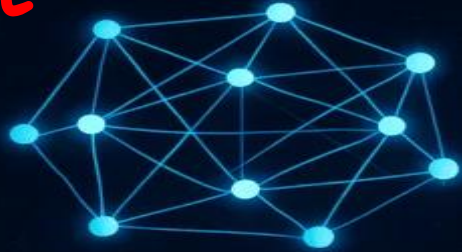
“Building Strong Foundations for Future AI Innovators”



Driving AI Evolution, Shaping Tomorrow's World

CORE MATHEMATICAL CONCEPTS BEHIND AI

LINEAR ALGEBRA



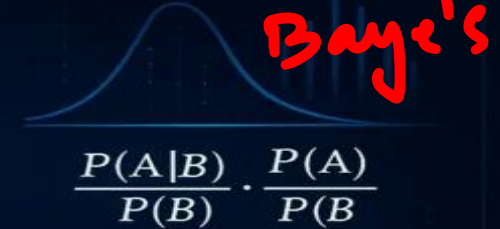
$$y = [X]$$

$$aIt/x - b$$

CALCULUS



PROBABILITY & STATISTICS



PROBABILITY & STATISTICS



$$P = \frac{\partial L / \partial w}{P/B}$$

= venn diagram -

DISCRETE MATH & GRAPH THEORY



$$G = (V, E)$$



* optimization }

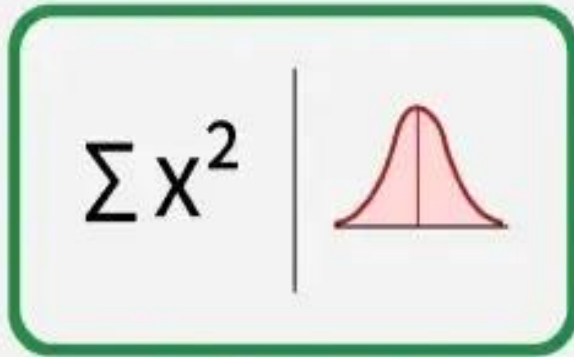


We Need Mathematics to make Computer Science Easy

Reasons

- Formal thinking and problem solving
- Algorithm design and analysis
- Cryptography and security
- Artificial intelligence and machine learning
- Data science and analytics

Time complexity
DSA problem



"Already done"

SYLLABUS DISCUSSION

Module 1: Linear Algebra and Vector Calculus

- Vectors and Matrices: Operations, properties, and applications in AI/ML.
- Eigenvalues and Eigenvectors: Principal Component Analysis (PCA).
- Vector Spaces: Concepts of basis, dimension, and linear independence.
- Matrix Decomposition: Singular Value Decomposition (SVD) and its applications.
- Gradient and Hessian: Introduction to vector calculus and its role in optimization

BASIC

Advanced

- saddle point



SYLLABUS DISCUSSION

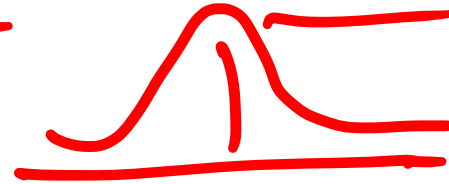
Module 2: Probability and Statistics

- Probability Theory: Basics of probability, conditional probability, and Bayes' theorem.

"Coin, dice, Card"

} Basic

- Random Variables and Distributions: Discrete and continuous distributions, expectation, and variance.



- Statistical Inference: Hypothesis testing, confidence intervals, and p-values.

estimation

- Regression Analysis: Linear regression, correlation, and causation.

curve fitting

- Bayesian Statistics: Introduction to Bayesian methods in AI/ML ✓

} Advanced



SYLLABUS DISCUSSION

Module 3: Optimization Techniques

- Convex Optimization: Convex sets, functions, and optimization problems.
- Gradient Descent: Optimization algorithms, learning rates, and convergence.
- Lagrange Multipliers: Constrained optimization techniques.
- Stochastic Optimization: Methods like Stochastic Gradient Descent (SGD).
- Optimization in Neural Networks: Backpropagation and cost function minimization

Advance
Topic



SYLLABUS DISCUSSION

Module 4: Discrete Mathematics and Graph Theory

- Set Theory: Basics of sets, functions, and relations.
- Combinatorics: Permutations, combinations, and their applications in algorithm analysis.
- Graph Theory: Graphs, trees, and networks; shortest path algorithms.
- Markov Chains: State transitions and applications in AI/ML.
- Discrete Probability: Probabilistic methods in discrete spaces



graph

Basic

Advanced

// Learn from past Experience //

Data

Analyze

What is Machine Learning?

ML = Machine + Learning



Training

Testing

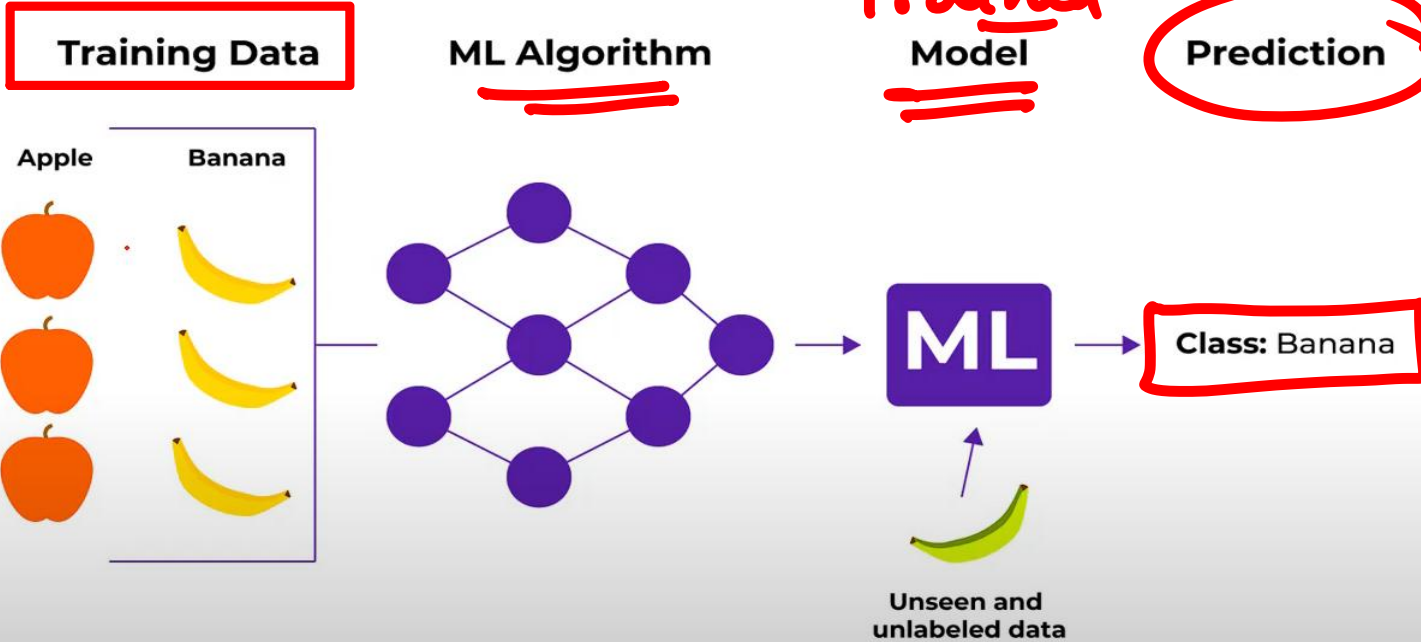


Create Artificial
Brain Using ML

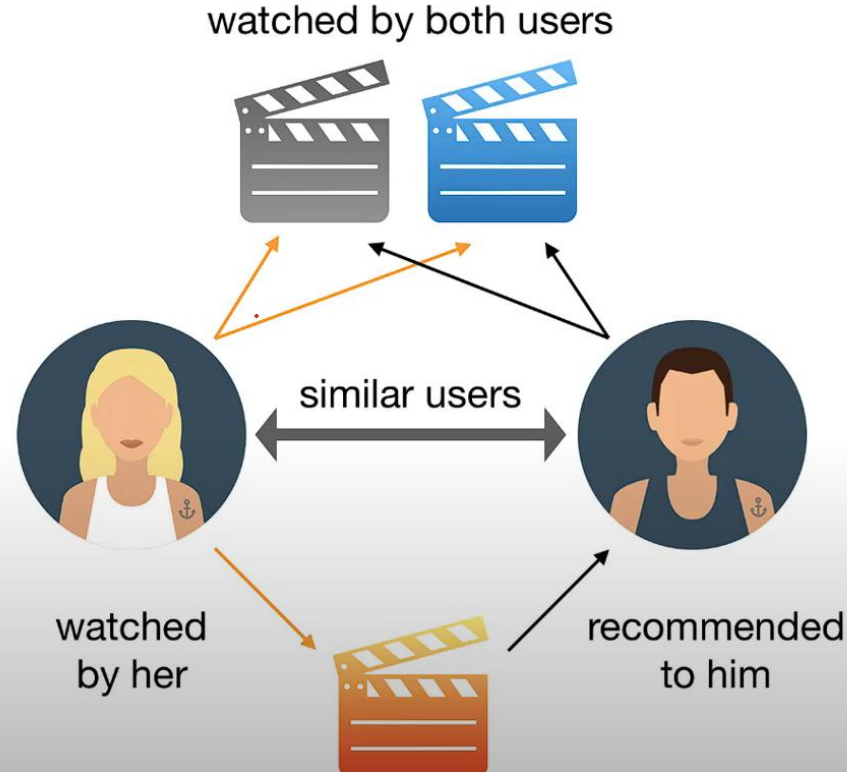
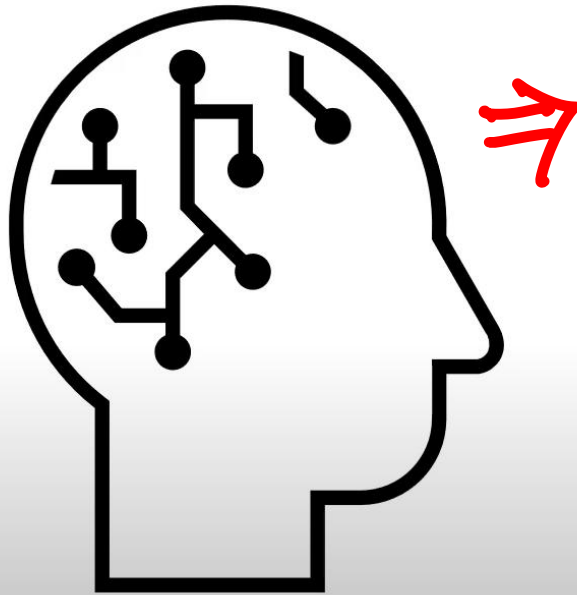
Able to
Make Prediction



Artificial Intelligence



Artificial Intelligence :-



chat Gpt, Perplexity etc ...



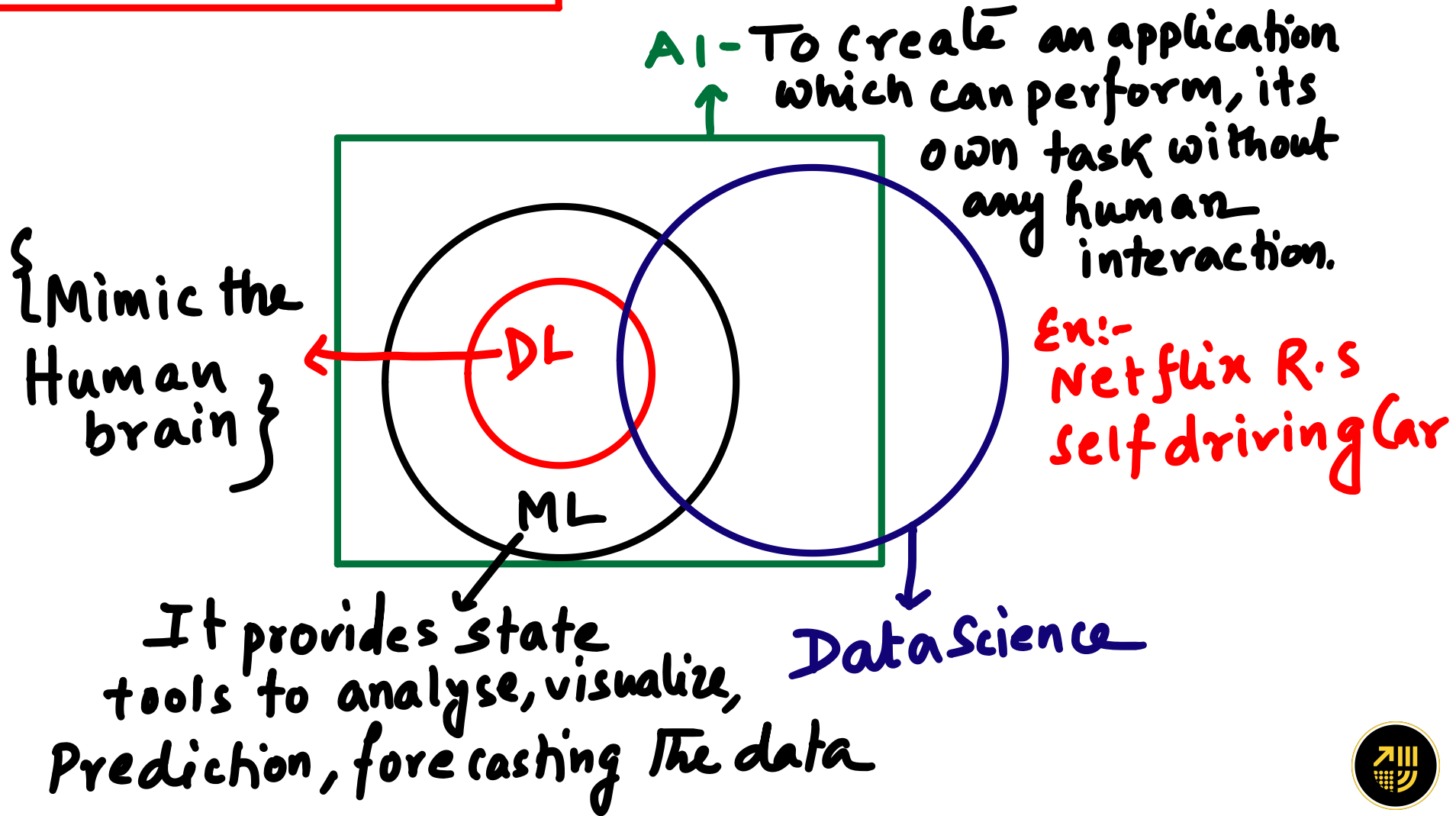
AI Chat Bot for your Business

IMPROVE YOUR
CUSTOMER SERVICE
WITH AI CHAT BOT.

"You Can..."



AI VS ML VS DL VS DS

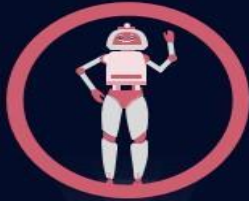


AI vs ML vs DL

Evolution →

Artificial Intelligence

Early artificial intelligence stirs excitement.



1950's

1960's

1970's

AI

Artificial Intelligence is the process of combining machines with human intelligence. It focuses on creating intelligent machines that can think and act like humans.

Machine Learning

Machine learning begins to flourish.



1980's

1990's

2000's

ML

Machine Learning is concerned with teaching machines to learn by training algorithms on vast amounts of data. It performs a task without the need for explicit instructions by using algorithms and statistical models

Deep Learning

Deep learning breakthroughs drive AI boom.



(NLP)

2010's

DL

Deep Learning is a kind of Machine Learning that focuses on data representations rather than task-specific methods. Deep Neural Networks are used based on the structure and function of the human brain

AI Areas of Research

Rule-based Reasoning

Machine Learning

Deep Learning

Natural Language Processing

Translation

Machine Vision

Speech to Text

Speech

Text to Speech

Robotics

Autonomous Vehicles

Artificial Intelligence

Artificial Intelligence (AI)

→ Application =

Machine Learning (ML)

(State tool)

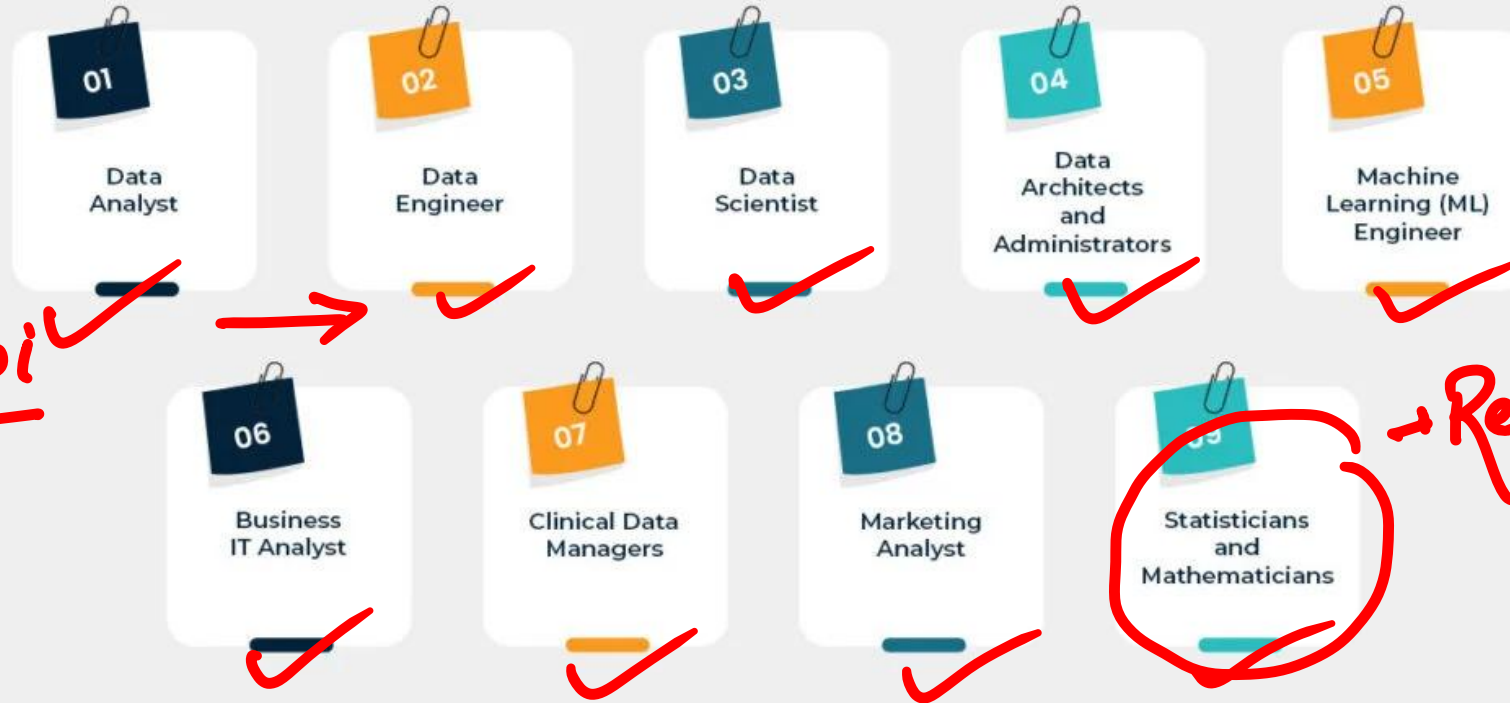
Deep Learning (DL)

(NLP)
(NN)

Data Science

Different Career Options In Data Science Domain

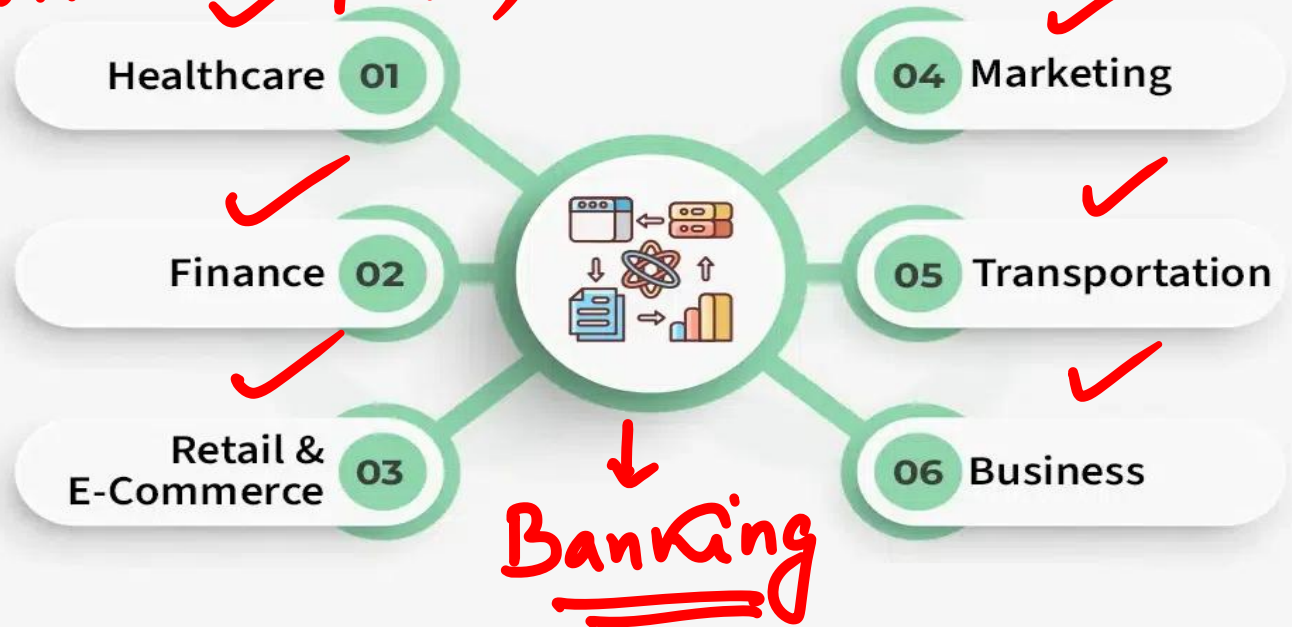
Python
Math
SQL
Excel
Powerbi



everyday life

The Role of Data Science in Everyday Life

(ultra Sonography)





Data Scientist Job Roles

Fresher

- ✓ Data Analyst
- ✓ Business Intelligence
- ✓ Data Science consultant
- ✓ Data Science Marketing Analyst

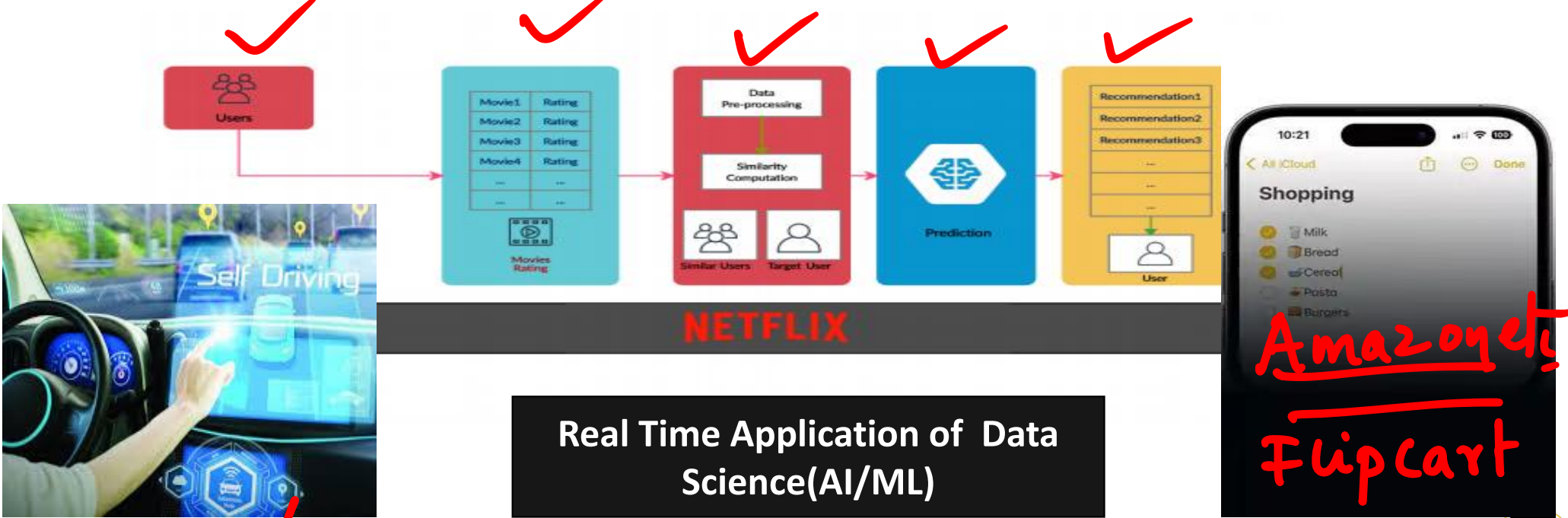
Intermediate

- Data Engineer
- Data Science Project Manager
- ML Engineer
- Data Science Ethicist

Experience

- ✓ Data Science Project Manager
- ✓ Data Scientist
- ✓ Data Science researcher





Tesla/Tata





“BE THE BEST VERSION OF YOURSELF”

